

Unique Equilibrium Can Be Cheaper than Unique Rationalizability in Team Contracting

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Abstract

Halac, Lipnowski, and Rappoport (2021) ask whether, when team production combines complementarities and substitutabilities, making full effort the unique Bayesian Nash equilibrium can cost less than making work every type's unique interim-correlated-rationalizable action. We answer affirmatively and compute both values. In a three-agent family, deterministic unit bonuses attain the equilibrium value 3, whereas the rationalizability infimum is $9(\alpha + \kappa)/(3\alpha + 2\kappa) > 3$. Both values optimize over all finite common-prior private-contract schemes with arbitrary type correlation, and three agents are minimal. The proof writes the two criteria as the same reciprocal-cost objective over nested marginal-product polytopes. A reciprocal-bonus potential yields the coalition polytope for equilibrium uniqueness; sequential deletion yields the smaller ranking polytope for rationalizability. The strict gap measures the incentive cost of replacing circular fixed-point exclusion by an acyclic deletion certificate.

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1 Introduction

A success-contingent team contract can support full effort while leaving the organization exposed to a coordination failure. If agents expect one another to shirk, a low-effort equilibrium may remain. A principal may therefore require full effort to be the unique equilibrium. When offers are private, she can also shape agents' higher-order beliefs about one another's incentives. Halac, Lipnowski, and Rappoport (2021) show that rank uncertainty can reduce the cost of unique implementation under a supermodular production technology.

For technologies with only complementarities, monotone-game arguments imply that unique Bayesian Nash equilibrium and unique interim correlated rationalizability lead to the same implementation value (Milgrom and Roberts, 1990; Halac, Lipnowski, and Rappoport, 2021). With only substitutabilities, full effort tests every agent at her least favorable opponent profile, so the positive perturbation in the implementation definition makes effort strictly dominant. Halac, Lipnowski, and Rappoport (2021, Section VI) isolate the unresolved mixed-sign case and ask whether unique-equilibrium implementation can then be strictly cheaper than unique-rationalizability implementation.

This paper gives an affirmative and exact answer. There are three agents. Agent 1 is a coordinator whose effort complements the efforts of agents 2 and 3, while the two specialists substitute for one another. For $\alpha, \kappa > 0$ and $\lambda \geq 2\kappa$, let

$$P(a_1, a_2, a_3) = \frac{\alpha a_1 + (\alpha + \lambda)(a_2 + a_3) + \kappa a_1(a_2 + a_3) - \lambda a_2 a_3}{3\alpha + \lambda + 2\kappa},$$

and let each effort cost equal $(\alpha + \kappa)/(3\alpha + \lambda + 2\kappa)$. The exact values are

$$\mathcal{V}_{\text{NE}} = 3, \quad \mathcal{V}_{\text{ICR}} = \frac{9(\alpha + \kappa)}{3\alpha + 2\kappa}.$$

Thus the cost gap is $3\kappa/(3\alpha + 2\kappa) > 0$. It holds throughout the open region $\alpha, \kappa > 0, \lambda > 2\kappa$, and also at its boundary $\lambda = 2\kappa$. The numerical instance $(\alpha, \kappa, \lambda) = (1, 1, 2)$ gives costs 3 and 18/5.

The equilibrium upper bound is the deterministic contract paying every agent one unit after success. The induced Bayesian game has no other pure or mixed equilibrium. The substantive issue is optimality: private information and correlated bonus types might, in principle, provide the same strategic assurance at lower expected cost. We rule this out without restricting the type spaces, their correlation, or the own-type-contingent bonuses.

The argument is organized by two convex sets of marginal-product budgets. The *coalition*

polytope records the inequalities implied by a weighted potential in the Bayesian agent normal form. If full effort is the unique equilibrium, then switching off any coalition must lower the potential. This bounds the coalition’s expected reciprocal bonuses. Cauchy’s inequality turns those bounds into a lower bound on expected transfers. The *ranking polytope* is the convex hull of the worst marginal-product vectors generated by agent rankings. The ranking formula described by Halac, Lipnowski, and Rappoport (2021, 2022) says that unique rationalizability minimizes the same reciprocal-cost objective over this smaller polytope. We give a self-contained proof specialized to team production.

The ranking polytope is always contained in the coalition polytope. The inclusion supplies a geometric comparison of the two implementation standards. For supermodular and submodular technologies, the two programs have the same minimum; this recovers the uniform-sign benchmarks. In the mixed-sign family above, the coalition optimum is the equal vector corresponding to unit bonuses, and the unit-bonus game is uniquely solvable. Every ranking vector, however, has a smaller fixed total marginal-product budget. The strict program gap is therefore exact, not merely a comparison between two proposed contracts.

The potential method is related to the weighted-potential contracting approach of Chan (2025) and to the potential-game framework of Monderer and Shapley (1996), but its role here is different. The potential is generated endogenously by type-specific reciprocal bonuses and is used as an impossibility certificate across arbitrary correlated private contracts; no potential-maximizer equilibrium-selection criterion is imposed. Recent work by Gossner and Veiel (2026) develops direct representations for interim correlated rationalizability in arbitrary finite games. Our ranking formula instead exploits the order structure of team production to reduce the contracting value itself to a finite polytope, while a shifted-rank construction keeps the approximating schemes finite. The result also complements the broader literatures on contracting for coordination (Ma, 1988; Winter, 2004; Halac, 2025), robust team incentives (Dai and Toikka, 2022), and information-based implementation (Oyama and Takahashi, 2020; Morris, Oyama, and Takahashi, 2024).

The economic distinction is between acyclic and circular reasoning. Unique rationalizability requires a deletion sequence: each agent must be induced to work when only previously certified agents are guaranteed to work. Unique equilibrium asks only whether the simultaneous best-response system has another fixed point. In the example, unequal specialist effort is self-defeating, while equal partial effort activates the coordinator and thereby destroys specialist indifference. This circular incompatibility excludes every alternative equilibrium but cannot be encoded by a ranking.

Section 2 presents the environment and solution concepts. Section 3 derives the two convex programs and their inclusion. Section 4 computes the exact separating family and proves minimality. Section 5 concludes.

2 Environment and implementation criteria

2.1 Team production and private contracts

Let $N = \{1, \dots, n\}$ be a finite set of agents. Agent i chooses $a_i \in \{0, 1\}$, where 1 denotes work, and

$$S(a) = \{i \in N : a_i = 1\}.$$

The project succeeds with probability $P(S(a))$, where $P : 2^N \rightarrow [0, 1]$ is strictly increasing. Thus

$$\Delta_i P(S) := P(S \cup \{i\}) - P(S) > 0 \quad (i \notin S).$$

Working costs agent i an amount $c_i > 0$.

An incentive scheme is a pair $\sigma = (q, B)$. Agent i has a finite type set T_i , and $q \in \Delta(T)$, $T = \prod_i T_i$, is a common prior whose marginal support is T_i . The principal privately informs each agent of her own type. Type t_i receives a nonnegative bonus $B_i(t_i)$ if the project succeeds and zero otherwise. Thus

$$u_i(a, t_i) = B_i(t_i)P(S(a)) - c_i a_i.$$

A type may encode information beyond the numerical bonus; no independence or symmetry restriction is imposed on q . Write q_i for the marginal distribution of T_i .

For $\varepsilon > 0$, let $\sigma^\varepsilon = (q, B + \varepsilon)$ denote the scheme obtained by adding ε to every type's success bonus. A behavioral strategy is a function $x_i : T_i \rightarrow [0, 1]$, interpreted as the probability of work. Conditional on the type profile, agents' behavioral randomizations are independent.

2.2 Interim correlated rationalizability

Following the rationalizability foundations of Bernheim (1984); Pearce (1984), we use the finite-game interim-correlated deletion procedure of Dekel, Fudenberg, and Morris (2007).

For every type t_i , initialize

$$R_i^0(t_i) = \{0, 1\}.$$

Given R^k , an action a_i belongs to $R_i^{k+1}(t_i)$ if it is a best response to some probability distribution β_i on $T_{-i} \times \{0, 1\}^{N \setminus \{i\}}$ such that

- (i) the marginal of β_i on T_{-i} is $q(\cdot \mid t_i)$; and
- (ii) β_i assigns probability one to opponent actions satisfying $a_j \in R_j^k(t_j)$ for every $j \neq i$.

The interim-correlated-rationalizable actions are

$$R_i^\infty(t_i) = \bigcap_{k \geq 0} R_i^k(t_i).$$

Finite type and action spaces imply finite stabilization. The conjecture β_i may correlate opponents' actions with their types and with one another; only its type marginal and surviving-action support are restricted.

Definition 1. A scheme σ *uniquely implements work in equilibrium* (is UIE) if, for every $\varepsilon > 0$, the all-work behavioral strategy profile is the unique Bayesian Nash equilibrium of σ^ε . It *uniquely implements work under rationalizability* (is UIR) if, for every $\varepsilon > 0$, work is the unique action in $R_i^\infty(t_i)$ for every i and t_i under σ^ε .

Equilibrium uniqueness is imposed over behavioral strategies, so the definition excludes nondegenerate mixed equilibria. The exact results below are unchanged if one instead requires uniqueness only among pure type-contingent strategies; see Remark 3.

The principal's expected payment at full effort is

$$W(\sigma) = P(N) \sum_{i \in N} \mathbb{E}[B_i(T_i)].$$

Define

$$\mathcal{V}_{\text{NE}} = \inf_{\sigma \text{ UIE}} W(\sigma), \quad \mathcal{V}_{\text{ICR}} = \inf_{\sigma \text{ UIR}} W(\sigma).$$

Every action assigned positive probability by a type in a Bayesian Nash equilibrium is interim correlated rationalizable. Hence every UIR scheme is UIE, and

$$\mathcal{V}_{\text{NE}} \leq \mathcal{V}_{\text{ICR}}. \tag{1}$$

For later reference, P is *supermodular* if $\Delta_i P(S) \leq \Delta_i P(H)$ whenever $S \subseteq H \subseteq N \setminus \{i\}$, and *submodular* if the reverse inequality holds.

3 Two marginal-product polytopes

This section expresses the two implementation requirements through the same strictly convex cost function on two nested feasible sets.

3.1 The coalition polytope

For nonempty $K \subseteq N$, define the loss from switching off precisely the agents in K at the full-effort profile:

$$d(K) = P(N) - P(N \setminus K).$$

Set $d(\emptyset) = 0$ and define

$$C(P) = \left\{ y \in \mathbb{R}_+^N : \sum_{i \in K} y_i \leq d(K) \text{ for every } K \subseteq N \right\}.$$

For $y \in \mathbb{R}_+^N$, let

$$\Psi_c(y) = \sum_{i \in N} \frac{c_i}{y_i},$$

with value $+\infty$ if some $y_i = 0$. Strict monotonicity of P makes $C(P)$ compact and gives it positive points, so Ψ_c has a unique minimizer on $C(P)$.

Proposition 2 (Reciprocal-bonus potential). *For every UIE scheme $\sigma = (q, B)$,*

$$y_i := c_i \mathbb{E} \left[\frac{1}{B_i(T_i)} \right] < \infty \tag{2}$$

belongs to $C(P)$, where $1/0 = +\infty$, and

$$W(\sigma) \geq P(N) \Psi_c(y). \tag{3}$$

Consequently,

$$\mathcal{V}_{\text{NE}} \geq P(N) \min_{y \in C(P)} \Psi_c(y). \tag{4}$$

Equality in (3) holds only if every $B_i(T_i)$ is constant almost surely.

Proof. Fix $\varepsilon > 0$. For a behavioral strategy profile x , let A denote the induced random

action profile and define

$$\Phi_\varepsilon(x) = \mathbb{E}[P(S(A))] - \sum_{i \in N} c_i \mathbb{E} \left[\frac{x_i(T_i)}{B_i(T_i) + \varepsilon} \right]. \quad (5)$$

Fix a type t_i and all strategy coordinates other than $x_i(t_i)$. The change in Φ_ε when this coordinate is changed from zero to one is

$$\begin{aligned} & \Phi_\varepsilon(1, x_{-(i,t_i)}) - \Phi_\varepsilon(0, x_{-(i,t_i)}) \\ &= \frac{q_i(t_i)}{B_i(t_i) + \varepsilon} ((B_i(t_i) + \varepsilon) \mathbb{E}[\Delta_i P(S(A_{-i})) \mid t_i] - c_i). \end{aligned} \quad (6)$$

The factor outside the parentheses is positive, and the expression in parentheses is the type's payoff difference between work and shirk. Continuity on the compact product of behavioral-strategy intervals guarantees a global maximizer. Every such maximizer satisfies the behavioral best-response conditions and is therefore a Bayesian Nash equilibrium. Since σ is UIE, full effort is the unique global maximizer.

Compare full effort with the pure type-contingent profile at which every type of every agent in a nonempty coalition K shirks and all other types work. Strict maximality gives

$$\sum_{i \in K} c_i \mathbb{E} \left[\frac{1}{B_i(T_i) + \varepsilon} \right] < d(K). \quad (7)$$

Letting $\varepsilon \downarrow 0$ and applying monotone convergence yields $y(K) \leq d(K)$ for every K . In particular, every reciprocal moment in (2) is finite.

Cauchy's inequality gives

$$\mathbb{E}[B_i(T_i)] \mathbb{E} \left[\frac{1}{B_i(T_i)} \right] \geq 1.$$

Since $y_i = c_i \mathbb{E}[1/B_i(T_i)]$,

$$\mathbb{E}[B_i(T_i)] \geq \frac{c_i}{y_i}.$$

Summing and multiplying by $P(N)$ proves (3) and (4). Equality in Cauchy's inequality holds only when $B_i(T_i)$ is constant almost surely. $\square$

Remark 3. The same lower bound holds if UIE is defined using only pure Bayesian equilibria. The potential in (5) is multilinear, so it has a pure global maximizer. Equation (6) makes

every pure global maximizer a pure Bayesian equilibrium. Pure-equilibrium uniqueness therefore identifies full effort as the global maximizer used in (7).

3.2 The ranking polytope

For $J \subseteq N \setminus \{i\}$, define the worst marginal product when the agents in J are known to work:

$$m_i(J) = \min_{J \subseteq H \subseteq N \setminus \{i\}} \Delta_i P(H). \quad (8)$$

Strict monotonicity gives $m_i(J) > 0$, and m_i is weakly increasing in J .

Let Π be the set of permutations, written as rank maps $\pi : N \rightarrow \{1, \dots, n\}$. Define

$$\text{Pred}_i(\pi) = \{j \in N : \pi(j) < \pi(i)\}, \quad m_i^\pi = m_i(\text{Pred}_i(\pi)),$$

and let

$$Q(P) = \text{co}\{m^\pi : \pi \in \Pi\} \subset \mathbb{R}_{++}^N.$$

Equivalently, if $\mu \in \Delta(\Pi)$ and

$$M_i(\mu) = \mathbb{E}_\mu[m_i^\pi],$$

then $Q(P) = \{M(\mu) : \mu \in \Delta(\Pi)\}$.

Proposition 4 (Ranking formula). *For every strictly increasing production technology,*

$$\mathcal{V}_{\text{ICR}} = P(N) \min_{y \in Q(P)} \Psi_c(y) = P(N) \min_{\mu \in \Delta(\Pi)} \sum_{i \in N} \frac{c_i}{M_i(\mu)}. \quad (9)$$

Proof. We first prove the lower bound. Fix a UIR scheme and $\varepsilon > 0$, and run the deletion procedure for σ^ε . Since work is the unique limiting action, work survives every round and shirk is deleted after finitely many rounds. Let

$$\tau_i(t_i) = \min\{k \geq 1 : 0 \notin R_i^k(t_i)\}.$$

At each type profile t in the support of q , order agents lexicographically by $(\tau_i(t_i), i)$, and denote the resulting permutation by $\pi^\varepsilon(t)$. Put

$$D_i(t) = \{j : \tau_j(t_j) < \tau_i(t_i)\}.$$

Fix i and t_i . For every t_{-i} in the conditional support, choose a set

$$H_i(t_{-i}; t_i) \in \arg \min_{D_i(t) \subseteq H \subseteq N \setminus \{i\}} \Delta_i P(H).$$

Immediately before shirk is deleted for type t_i , every agent in $D_i(t)$ is forced to work at type profile t , while both actions remain available to every opponent whose deletion time is weakly larger. Hence, at each conditional type profile, the conjecture that assigns probability $q(t_{-i} \mid t_i)$ to the action profile with work set $H_i(t_{-i}; t_i)$ is admissible. Shirk is not a best response to this conjecture, so

$$(B_i(t_i) + \varepsilon) \mathbb{E}[m_i(D_i(T)) \mid T_i = t_i] > c_i. \quad (10)$$

Combining (10) with the fact that tie breaking can only add agents to the predecessor set, $D_i(t) \subseteq \text{Pred}_i(\pi^\varepsilon(t))$, and using monotonicity of m_i gives

$$B_i(t_i) + \varepsilon > \frac{c_i}{\mathbb{E}[m_i^{\pi^\varepsilon(T)} \mid T_i = t_i]}. \quad (11)$$

Let μ^ε be the distribution of $\pi^\varepsilon(T)$. Taking expectations in (11) and applying Jensen's inequality to $z \mapsto 1/z$ gives

$$\mathbb{E}[B_i(T_i)] + \varepsilon > \frac{c_i}{M_i(\mu^\varepsilon)}.$$

Therefore

$$\frac{W(\sigma)}{P(N)} + n\varepsilon > \sum_i \frac{c_i}{M_i(\mu^\varepsilon)} \geq \min_{\mu \in \Delta(\Pi)} \sum_i \frac{c_i}{M_i(\mu)}.$$

Letting $\varepsilon \downarrow 0$ proves the lower bound.

For the reverse inequality, fix $\mu \in \Delta(\Pi)$ and an integer $L \geq n$. Draw $\pi \sim \mu$ and, independently, draw s uniformly from $\{0, \dots, L-1\}$. Agent i observes the label

$$T_i = s + \pi(i),$$

and T_i is taken to have its marginal support as type space. Realized labels are distinct and rank agents in the same order as π . For every type k , set

$$B_i^L(k) = \frac{c_i}{\mathbb{E}[m_i^\pi \mid T_i = k]}. \quad (12)$$

Fix a positive perturbation. By induction on the deletion rounds, work survives for every type. Indeed, provided that opponents' work actions survived the preceding round, a type can conjecture that all opponents work, and

$$\Delta_i P(N \setminus \{i\}) \geq \mathbb{E}[m_i^\pi \mid T_i = k],$$

so (12) and the perturbation make work strictly optimal under that conjecture. This starts the induction at round zero and propagates it to every later round.

Now delete shirk in increasing order of the numerical labels. Suppose shirk has already been deleted for every label below k . Conditional on $T_i = k$, every predecessor in the realized ranking has a lower label and is therefore forced to work. Whatever surviving actions are assigned to higher-label opponents, the statewise marginal product is at least m_i^π . Conditional expectation and (12) make work strictly better than shirk. Induction over the finitely many labels deletes shirk for every type, so the scheme is UIR.

For an interior label $k \in \{n, \dots, L\}$, every ranking in the support of μ is compatible with $T_i = k$, with the unique shift $s = k - \pi(i)$, and the conditional distribution of π is exactly μ . Hence

$$B_i^L(k) = \frac{c_i}{M_i(\mu)}.$$

Conditional on any realized rank, the probability of a boundary label is exactly $(n - 1)/L$. Since the finitely many numbers $m_i(J)$ are bounded away from zero, boundary bonuses are uniformly bounded in L . It follows that

$$\lim_{L \rightarrow \infty} \mathbb{E}[B_i^L(T_i)] = \frac{c_i}{M_i(\mu)}.$$

Thus finite UIR schemes have costs converging to the objective associated with μ . Minimizing over μ completes the proof. $\square$

The value in (9) is an infimum over finite schemes and a minimum over the finite-dimensional ranking simplex. The shifted-rank construction proves approximation; finite attainment of the infimum is not asserted.

3.3 The nested-program comparison

Lemma 5 (Ranking budgets satisfy every coalition inequality). *For every strictly increasing P ,*

$$Q(P) \subseteq C(P).$$

Proof. It is enough to consider a ranking vector m^π . Fix $K \subseteq N$, and list its members as $i_1, \dots, i_{|K|}$ in their order under π . Put

$$H_\ell = (N \setminus K) \cup \{i_1, \dots, i_{\ell-1}\}.$$

Every predecessor of i_ℓ lies in H_ℓ , so the definition of the worst marginal product gives

$$m_{i_\ell}^\pi \leq \Delta_{i_\ell} P(H_\ell).$$

Summing along the chain that adds the agents in K yields

$$\sum_{i \in K} m_i^\pi \leq \sum_{\ell=1}^{|K|} \Delta_{i_\ell} P(H_\ell) = P(N) - P(N \setminus K) = d(K).$$

Hence $m^\pi \in C(P)$; convexity gives the result. $\square$

Theorem 6 (Program comparison and exactness certificate). *For every strictly increasing technology,*

$$P(N) \min_{y \in C(P)} \Psi_c(y) \leq \mathcal{V}_{\text{NE}} \leq \mathcal{V}_{\text{ICR}} = P(N) \min_{y \in Q(P)} \Psi_c(y), \quad (13)$$

and $Q(P) \subseteq C(P)$.

Let y^C be the unique minimizer of Ψ_c on $C(P)$. If the deterministic bonuses

$$B_i^C = \frac{c_i}{y_i^C} \quad (14)$$

uniquely implement work in equilibrium, then

$$\mathcal{V}_{\text{NE}} = P(N) \Psi_c(y^C). \quad (15)$$

Moreover, every UIE scheme attaining this value pays B_i^C to agent i almost surely.

Proof. The chain in (13) follows from Proposition 2, (1), and Proposition 4; the set inclusion is Lemma 5. If the contract in (14) is UIE, its cost is the leftmost term in (13), proving (15).

If another scheme attains that value, the two nonnegative slacks in

$$\frac{W(\sigma)}{P(N)} - \Psi_c(y) \quad \text{and} \quad \Psi_c(y) - \Psi_c(y^C)$$

must both vanish. Strict convexity of Ψ_c gives $y = y^C$. The first equality is the sum of the agents' reciprocal-moment inequalities; hence every agent's nonnegative slack vanishes separately. Equality in Cauchy's inequality makes each bonus constant almost surely, and the constant is c_i/y_i^C . $\square$

The theorem separates two questions. The coalition program is a universal lower bound on unique-equilibrium cost; checking the deterministic contract in (14) establishes exactness. Unique rationalizability is already represented exactly by the smaller ranking polytope.

Corollary 7 (Uniform-sign technologies). *If P is supermodular or submodular, then*

$$\min_{y \in C(P)} \Psi_c(y) = \min_{y \in Q(P)} \Psi_c(y), \quad \mathcal{V}_{\text{NE}} = \mathcal{V}_{\text{ICR}}. \quad (16)$$

Proof. Suppose first that P is submodular. Then the minimum in (8) is always attained at $N \setminus \{i\}$, so every ranking vector equals

$$\bar{y}_i = \Delta_i P(N \setminus \{i\}) = d(\{i\}).$$

Thus $Q(P) = \{\bar{y}\}$. By Lemma 5, $\bar{y} \in C(P)$, while the singleton constraints defining $C(P)$ give $y_i \leq \bar{y}_i$ for every feasible y . Since Ψ_c is decreasing in every coordinate, $\bar{y}$ minimizes it on both sets.

Suppose next that P is supermodular. Then $m_i(J) = \Delta_i P(J)$. The deficit function d is normalized, increasing, and submodular. Let

$$\mathcal{B}(d) = \{y \in C(P) : y(N) = d(N)\}$$

be its base polytope. For any ranking π , reverse the ranking and apply the greedy increments of d . The increment assigned to agent i is

$$P(\text{Pred}_i(\pi) \cup \{i\}) - P(\text{Pred}_i(\pi)) = m_i^\pi.$$

The greedy support-function argument for submodular base polytopes (Edmonds, 1970) shows that the convex hull of these vectors is exactly $\mathcal{B}(d)$. Indeed, after ordering arbitrary

weights $w_{\sigma(1)} \geq \dots \geq w_{\sigma(n)}$, summation by parts bounds $w \cdot y$ over $\mathcal{B}(d)$ by replacing every prefix sum $y(\{\sigma(1), \dots, \sigma(k)\})$ with the corresponding value of d ; the reverse-ranking greedy vector attains every bound. Hence

$$Q(P) = \mathcal{B}(d).$$

Finally, every $y \in C(P)$ is coordinatewise dominated by some $z \in \mathcal{B}(d)$. To see this, maximize $z(N)$ over the compact set $\{z \in C(P) : z \geq y\}$. If the maximizer had $z(N) < d(N)$, then inability to increase coordinate i would provide a tight set $K_i \ni i$. If A and B are tight, then

$$d(A) + d(B) = z(A) + z(B) = z(A \cup B) + z(A \cap B) \leq d(A \cup B) + d(A \cap B) \leq d(A) + d(B),$$

so every inequality is an equality and $A \cup B$ is tight. Iterating this closure property, $N = \bigcup_i K_i$ would be tight, a contradiction. Thus $z(N) = d(N)$. Since Ψ_c is decreasing coordinatewise, its minimum on $C(P)$ is attained on $\mathcal{B}(d) = Q(P)$. The value equality in (16) now follows from Theorem 6. $\square$

4 An exact and minimal separation

4.1 The mixed-sign family

Fix

$$\alpha > 0, \quad \kappa > 0, \quad \lambda \geq 2\kappa, \tag{17}$$

and put $Z = 3\alpha + \lambda + 2\kappa$. Define

$$ZP(a_1, a_2, a_3) = \alpha a_1 + (\alpha + \lambda)(a_2 + a_3) + \kappa a_1(a_2 + a_3) - \lambda a_2 a_3, \tag{18}$$

with common effort cost

$$c_1 = c_2 = c_3 = c := \frac{\alpha + \kappa}{Z}. \tag{19}$$

Theorem 8 (Exact and minimal separation). *For every parameter triple satisfying (17), the technology in (18) is strictly increasing and is neither supermodular nor submodular. Its*

exact implementation values are

$$\mathcal{V}_{\text{NE}} = 3, \quad \mathcal{V}_{\text{ICR}} = \frac{9(\alpha + \kappa)}{3\alpha + 2\kappa} > 3. \quad (20)$$

The equilibrium value is attained by deterministic unit bonuses, and every UIE scheme attaining it pays bonus 1 to every agent almost surely. No strictly increasing technology with fewer than three agents admits a strict gap between the two values.

The proof occupies the next three subsections. The rationalizability value in (20) is an infimum over finite schemes. The shifted-rank schemes in the proof of Proposition 4 approach it; finite attainment is neither asserted nor required.

The normalized marginal products are

$$Z\Delta_1 P(a_2, a_3) = \alpha + \kappa(a_2 + a_3), \quad (21)$$

$$Z\Delta_2 P(a_1, a_3) = \alpha + \lambda + \kappa a_1 - \lambda a_3, \quad (22)$$

$$Z\Delta_3 P(a_1, a_2) = \alpha + \lambda + \kappa a_1 - \lambda a_2. \quad (23)$$

All are positive. Since $P(\emptyset) = 0$ and $P(N) = 1$, strict monotonicity also gives $P(S) \in [0, 1]$. The interactions between agent 1 and each specialist equal $\kappa/Z > 0$, while the interaction between the specialists equals $-\lambda/Z < 0$. Thus the technology has both signs.

4.2 The coalition optimum and unit bonuses

Lemma 9 (Coalition program). *For the family in (18)–(19), the unique minimizer of Ψ_c on $C(P)$ is*

$$y^C = (c, c, c), \quad (24)$$

and the minimum is 3.

Proof. Write $r_i = y_i/c$. The coalition constraints for $K = \{2\}, \{3\}, \{1, 2\}, \{1, 3\}$, and $\{1\}$ imply

$$r_2 \leq 1, \quad r_3 \leq 1, \quad r_1 + r_2 \leq 2, \quad r_1 + r_3 \leq 2, \quad r_1 \leq \frac{\alpha + 2\kappa}{\alpha + \kappa} < 2. \quad (25)$$

If $r_1 \leq 1$, then $r_i \leq 1$ for all i , and $\sum_i 1/r_i \geq 3$. If $1 < r_1 < 2$, the pairwise constraints give $r_2, r_3 \leq 2 - r_1$, so

$$\sum_i \frac{1}{r_i} \geq \frac{1}{r_1} + \frac{2}{2 - r_1},$$

and

$$\frac{1}{r_1} + \frac{2}{2-r_1} - 3 = \frac{(r_1-1)(3r_1-2)}{r_1(2-r_1)} > 0.$$

Equality is therefore possible only at $r = (1, 1, 1)$.

This vector is feasible. The five constraints displayed in (25) hold, while the two remaining coalition constraints reduce to

$$2(\alpha + \kappa) \leq 2\alpha + 2\kappa + \lambda, \quad 3(\alpha + \kappa) \leq 3\alpha + 2\kappa + \lambda,$$

which follow from $\lambda \geq 2\kappa$. Since $\Psi_c(y) = \sum_i 1/r_i$, the result follows. $\square$

Lemma 10 (Unique mixed equilibrium). *The deterministic contract $B_1 = B_2 = B_3 = 1$ uniquely implements work in equilibrium.*

Proof. Fix $\varepsilon > 0$, and let $x_i \in [0, 1]$ be agent i 's work probability. Multiplying each payoff difference by $Z/(1 + \varepsilon) > 0$, its sign is the sign of

$$\begin{aligned} g_1(x) &= \alpha + \kappa(x_2 + x_3) - \rho, \\ g_2(x) &= \alpha + \lambda + \kappa x_1 - \lambda x_3 - \rho, \\ g_3(x) &= \alpha + \lambda + \kappa x_1 - \lambda x_2 - \rho, \end{aligned}$$

where

$$\rho = \frac{\alpha + \kappa}{1 + \varepsilon} < \alpha + \kappa. \quad (26)$$

At equilibrium, $x_i = 0$ requires $g_i \leq 0$, $0 < x_i < 1$ requires $g_i = 0$, and $x_i = 1$ requires $g_i \geq 0$.

Suppose $x_2 > x_3$. Then

$$g_2 - g_3 = \lambda(x_2 - x_3) > 0.$$

If $x_2 < 1$, agent 2 is interior, so $g_2 = 0$ and $g_3 < 0$, forcing $x_3 = 0$. But then

$$g_2 = \alpha + \lambda + \kappa x_1 - \rho > \lambda - \kappa \geq \kappa > 0,$$

a contradiction. Hence $x_2 = 1$. If $x_3 > 0$, agent 3 is interior and $g_3 = 0$. Yet $g_1 > 0$, so $x_1 = 1$, after which $g_3 = \alpha + \kappa - \rho > 0$, again a contradiction. Thus $x_3 = 0$. This gives $g_1 = \alpha + \kappa - \rho > 0$, hence $x_1 = 1$, and then $g_3 > 0$, contradicting $x_3 = 0$. Therefore $x_2 > x_3$ is impossible. By symmetry, $x_3 > x_2$ is impossible as well.

Write $x_2 = x_3 = y$. If $y = 0$, then

$$g_2 = \alpha + \lambda + \kappa x_1 - \rho > \lambda - \kappa \geq \kappa,$$

contradicting $x_2 = 0$. If $0 < y < 1$, specialist indifference gives

$$\lambda y = \alpha + \lambda + \kappa x_1 - \rho > \lambda - \kappa,$$

so $y > 1 - \kappa/\lambda \geq 1/2$. Consequently,

$$g_1 = \alpha + 2\kappa y - \rho > \kappa(2y - 1) > 0,$$

and $x_1 = 1$. Substitution into the specialist indifference equation yields

$$\lambda y = \alpha + \lambda + \kappa - \rho > \lambda,$$

contradicting $y < 1$. Hence $y = 1$, and then $g_1 > 0$ forces $x_1 = 1$. At full effort, all three payoff differences are strictly positive. Full effort is therefore the unique equilibrium, including among mixed strategies. $\square$

By (24) and Lemma 10, the contract prescribed by the exactness certificate (14) is precisely the unit-bonus contract. Theorem 6 now gives

$$\mathcal{V}_{\text{NE}} = 3,$$

and every cost-minimizing UIE scheme pays one almost surely to each agent.

Remark 11 (Sharpness for unit bonuses). The threshold $\lambda \geq 2\kappa$ is sharp for the unit-bonus contract. If $0 < \lambda < 2\kappa$, define

$$y_\varepsilon = \frac{\rho - \alpha}{2\kappa}, \quad x_{1,\varepsilon} = \frac{\rho - \alpha - \lambda + \lambda y_\varepsilon}{\kappa}, \quad x_{2,\varepsilon} = x_{3,\varepsilon} = y_\varepsilon,$$

where ρ is given by (26). For every sufficiently small $\varepsilon > 0$, all coordinates lie in $(0, 1)$ and satisfy $g_1 = g_2 = g_3 = 0$. Hence they form a nondegenerate mixed equilibrium. Moreover,

$$(x_{1,\varepsilon}, x_{2,\varepsilon}, x_{3,\varepsilon}) \longrightarrow \left(1 - \frac{\lambda}{2\kappa}, \frac{1}{2}, \frac{1}{2}\right) \quad \text{as } \varepsilon \downarrow 0.$$

The remark concerns the unit-bonus construction; it does not claim that 2κ is a necessary

threshold for every possible private scheme.

4.3 The ranking optimum

Lemma 12 (Ranking program). *For the family in (18)–(19),*

$$\min_{y \in Q(P)} \Psi_c(y) = \frac{9(\alpha + \kappa)}{3\alpha + 2\kappa}. \quad (27)$$

Proof. Write $\widehat{m}_i = Zm_i$. Equations (21)–(23) give

$$\begin{aligned} \widehat{m}_1(J) &= \alpha + \kappa|J|, \\ \widehat{m}_2(J) &= \alpha + \kappa \mathbf{1}_{\{1 \in J\}}, \\ \widehat{m}_3(J) &= \alpha + \kappa \mathbf{1}_{\{1 \in J\}}. \end{aligned}$$

The six rankings generate four distinct vectors:

Order from first to last	$(\widehat{m}_1^\pi, \widehat{m}_2^\pi, \widehat{m}_3^\pi)$
(1, 2, 3) or (1, 3, 2)	$(\alpha, \alpha + \kappa, \alpha + \kappa)$
(2, 1, 3)	$(\alpha + \kappa, \alpha, \alpha + \kappa)$
(3, 1, 2)	$(\alpha + \kappa, \alpha + \kappa, \alpha)$
(2, 3, 1) or (3, 2, 1)	$(\alpha + 2\kappa, \alpha, \alpha)$

Every row has coordinate sum $3\alpha + 2\kappa$. If $d_i = Zy_i$ for $y \in Q(P)$, then

$$\Psi_c(y) = (\alpha + \kappa) \sum_{i=1}^3 \frac{1}{d_i}, \quad \sum_{i=1}^3 d_i = 3\alpha + 2\kappa.$$

Cauchy's inequality yields

$$\Psi_c(y) \geq \frac{9(\alpha + \kappa)}{3\alpha + 2\kappa}.$$

Equality is feasible: assign probability $1/3$ to each of the rankings

$$(1, 2, 3), \quad (2, 1, 3), \quad (3, 1, 2).$$

Their expected normalized coordinates are all $(3\alpha + 2\kappa)/3$, proving (27). $\square$

By Proposition 4 and Lemma 12, the second equality in (20) holds. Subtracting the

equilibrium value gives

$$\mathcal{V}_{\text{ICR}} - \mathcal{V}_{\text{NE}} = \frac{3\kappa}{3\alpha + 2\kappa} > 0. \quad (28)$$

Corollary 13 (Numerical instance). *For $(\alpha, \kappa, \lambda) = (1, 1, 2)$,*

$$P(a_1, a_2, a_3) = \frac{a_1 + 3a_2 + 3a_3 + a_1a_2 + a_1a_3 - 2a_2a_3}{7}, \quad c_i = \frac{2}{7},$$

and

$$\mathcal{V}_{\text{NE}} = 3, \quad \mathcal{V}_{\text{ICR}} = \frac{18}{5}.$$

4.4 Minimality and mechanism

With two agents there is only one cross-difference,

$$P(\{1, 2\}) - P(\{1\}) - P(\{2\}) + P(\emptyset).$$

Its sign makes every strictly increasing two-agent technology supermodular or submodular. Corollary 7 then gives $\mathcal{V}_{\text{NE}} = \mathcal{V}_{\text{ICR}}$, so three agents are necessary.

Proof of Theorem 8. Strict monotonicity and the two signs of interaction follow from (21)–(23). Lemmas 9 and 10 and Theorem 6 give $\mathcal{V}_{\text{NE}} = 3$ and the almost-sure uniqueness of the cost-minimizing bonuses. Proposition 4 and Lemma 12 give the stated rationalizability value and hence the strict gap. The preceding two-agent argument establishes minimality. $\square$

The two polytopes reveal the source of the gap. The coalition optimizer satisfies

$$Zy^C = (\alpha + \kappa, \alpha + \kappa, \alpha + \kappa), \quad Zy^C(N) = 3\alpha + 3\kappa.$$

Every point of the ranking polytope instead satisfies

$$Zy(N) = 3\alpha + 2\kappa.$$

Thus the equal coalition optimum lies outside the ranking polytope by the exact aggregate amount κ/Z . The strictly convex reciprocal objective transforms this missing marginal-product budget into the cost gap in (28).

The best-response argument explains why the coalition optimum is nevertheless attainable under equilibrium uniqueness. If the specialists use unequal effort probabilities, the one

working more has the stronger incentive to work, which is incompatible with equilibrium mixing or boundary behavior. If they use the same partial effort probability, the coordinator strictly prefers to work; once the coordinator works, specialist indifference is impossible. This is a circular exclusion of alternative fixed points. A rationalizability proof cannot use the circle: at each deletion round it must certify one agent using only agents certified earlier.

The interaction parameters have distinct functions. Complementarity κ determines the absolute gap and the relative saving

$$\frac{\mathcal{V}_{\text{ICR}} - \mathcal{V}_{\text{NE}}}{\mathcal{V}_{\text{ICR}}} = \frac{\kappa}{3(\alpha + \kappa)}.$$

Substitutability λ eliminates alternative equilibria but does not enter the rationalizability value: a specialist's worst-case conjecture always assigns work to the other specialist, canceling the term λ from her worst marginal product.

5 Conclusion

When a team technology combines complements and substitutes, equilibrium uniqueness and unique rationalizability can impose genuinely different incentive costs. The distinction is exact in a minimal three-agent environment. Unit bonuses minimize the cost of unique equilibrium even over arbitrary correlated private-contract schemes, while every scheme that uniquely rationalizes work costs strictly more.

The comparison rests on a common objective over nested marginal-product polytopes. Global potential comparisons force the coalition polytope, whereas sequential deletion forces the ranking polytope. Uniform-sign technologies have the same optimum over the two sets. Mixed signs can separate them, and equilibrium uniqueness is cheaper when a circular best-response structure implements an optimizer that no acyclic ranking can certify. The next question is to characterize the signed interaction structures for which this strict geometric separation occurs.

Data availability

The article uses no data. All results are analytic; no computational replication files are required.

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